Mowaz Mohammed Abdul Karim

254, West Shantibagh, Dhaka 1217, Bangladesh

Email: mowaz.mohd@gmail.com | LinkedIn | Google Scholar | Portfolio

Phone: +8801305-685744 (Primary) | +8801521-562463 (WhatsApp)

PROFESSIONAL SUMMARY

A diligent Civil and Environmental Engineer and research-driven problem solver committed to delivering technically sound solutions that drive real-world impact.

EDUCATION

B.Sc in Civil and Environmental Engineering *CGPA: 3.12*

Shahjalal University of Science and Technology, Sylhet, Bangladesh

WORK EXPERIENCE

- **Research Associate, SUST Research Center, SUST** *July 2025 - Present*

Funded by University Grant Commission; Project ID: AS/2025/1/19 *Progress 80%*

Project Investigators: Dr. Shilpy Rani Basak (PI); Nusrat Jahan Ekra (CI).

Key Responsibilities:

Designing research framework, preparing comprehensive proposals and reports, conducting qualitative and quantitative surveys, analyzing data using SPSS, performing FEA simulations in ANSYS, and developing machine learning models for predictive design optimization.

- **Research Assistant, Department of Civil and Environmental Engineering, SUST** *January – July 2025*

Supervisor: Dr. Md. Imran Kabir

Designed an innovative mechanism to harness renewable energy from human footsteps by utilizing crowd spaces and the high population density of Bangladesh.

SKILLS

Languages: C, C++, Python, Matlab.

Software: SWMM, EPANET, AutoCAD, Fusion 360, ArcGIS, QGIS, Ansys, SPSS, Google Earth Engine.

Tools: Microsoft Office Suite, Google Suite, Microsoft Azure, KoboToolbox, Adobe Illustrator, VS Code.

Industry: Water Quality Assessments, Pollution Control, EIA, Hydrodynamics and Hydraulics, Artificial Intelligence.

RESEARCH EXPERIENCE

- **Prediction of Seismic Effect on Structure using Deep Learning [Under Review]** *Undergraduate Thesis*

Developed deep learning models comprising cGAN and ANN architectures to predict von Mises stress plots and maximum stress values of structural elements under vertical and lateral loading.

- **Plaster Surface Crack Recovery Optimization via Sustainable Spray Based Microbial-Induced Calcite Precipitation [Under Review]**

In this work we investigated the use of spray-based Microbial Induced Calcite Precipitation (MICP) as a plaster crack remediation method compared to conventional approaches.

- **Water Sanitation and Hygiene Survey of Basirpur Village in Sylhet City [Ongoing]**

Generated a comprehensive statistical report combining questionnaire surveys with field and laboratory tests to analyze iron, arsenic, and fecal coliform concentrations, assessing sanitation and hygiene practices across households.

- **A novel archetype of energy generation from human footsteps. [Ongoing]**

Introduced an energy harvesting system that converts compression to energy using fluid dynamics method.

- **Public Perception on Quarantine during the COVID-19 Outbreak in Bangladesh: A Community Survey-based Study**

Doi: 10.29333/jcei/11703

Conducted statistical analysis based on public survey report to provide outcomes of public perception on COVID-19 outbreak in the community.

- **Archetype of energy generation system from concentrated solar power and thermal gradient of water bodies and atmosphere**

Doi: 10.1109/ICDRET60388.2024.10503636

Presented a methodology to harvest energy from thermal gradient caused by concentrated sunlight using the temperature difference.

- **An IoT Based Smart Vault Security and Monitoring System with Zero UI**

Doi: 10.1109/ICREST57604.2023.10070057

Proposed a security vault system with three level security that functions without any physical contact.

PROJECTS

- **VisionX** *2025*

Built AI-powered smart goggles to enable blind individuals to read texts, identify objects and currency, and manage small business through voice control.

- **MedPunctual** *2024*

Developed a pill box designed to ensure timely medication delivery, daily health monitoring, and pill supply management using computer vision.

- **I-Braille** *2022*

Designed a low-cost portable Braille display focusing on user ease, enabling blind and deaf individuals to read digital and physical text.

ACHIEVEMENTS

- MIT Solve 2025: Featured on MIT Solve website for the project of VisionX.
- Microsoft Imagine Cup 2024: Qualified for third phase for the project of MedPunctual.
- University Innovation Hub Program SUST-IC3 (Runner-Up) – Gathered 60K pre-seed fund for VisionX from World Bank.
- Pitch Propel Prosper (2nd Runner-Up): Qualified as top 3 business ideas for the project VisionX.
- Battle of Minds 2023 (Bootcamp): Qualified for third phase of Battle of Minds 2023 case competition.
- IC4IR 2021: Idea Contest (Champion) – Gathered 1M BDT as pre-seed fund for the I-Braille Project.
- IEEE Satellite Expedition Contest 2023 (Runner-Up) – Proposing a satellite design for tracing pollutant origins in the atmosphere.
- bdSTEM Competition 2021: National STEM Competition (Finalist) – Energy production from CSP.
- Mechnovation 2022: Robonix LFR competition (Runner-Up) – Built a line follower bot for an obstacle course.
- IEEE YESIST 12-2021 WePOWER Track (Finalist) – Developed a method for energy generation from footstep.
- International Poster Competition organized by UniV (Gold Award) – For the research work of energy generation.

LEADERSHIP EXPERIENCE

- **Graduate Mentor, ACI SUST Student Chapter, Shahjalal University of Science and Technology**

Instructed undergraduate students about technical writing and presentation. Directed fifteen students individually for competitions.

- **General Secretary, AAJ Muktomancho, Shahjalal University of Science and Technology**

Organized 4 events and 2 workshops for performing and visual arts. Performed as an artist and served as an instructor of mime arts.

- **Joint Secretary, RoboSUST, Shahjalal University of Science and Technology**

Arranged 3 workshops and 2 events. Instructed basic robotics for high schools twice. Developed 4 robotics projects.

COURSES

- **Build Basic Generative Adversarial Networks (GANs)** *deeplearning.ai on Coursera*
- **Supervised Machine Learning: Regression and Classification** *deeplearning.ai on Coursera*
- **Neural Networks and Deep Learning** *deeplearning.ai on Coursera*
- **Python for Data Science, AI & Development** *IBM on Coursera*
- **Remote Sensing Image Acquisition, Analysis and Applications** *UNSW, Sydney on Coursera*

REFERENCE

Dr. Mushtaq Ahmed

Professor

Dept. of Civil and Environmental Engineering,

Shahjalal University of Science and Technology

Email: mushtaq_cee@yahoo.com

Dr. Md. Imran Kabir

Professor

Dept. of Civil and Environmental Engineering,

Shahjalal University of Science and Technology

Email: imran.kabir-cee@sust.edu

Dr. Ahmad Hasan Nury

Professor

Dept. of Civil and Environmental Engineering,

Shahjalal University of Science and Technology

Email: hasan-cee@sust.edu